CS & IT ENGINEERING

Discrete Mathematics

Set Theory and Algebra

DPP- 05 Discussion Notes



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[MCQ]

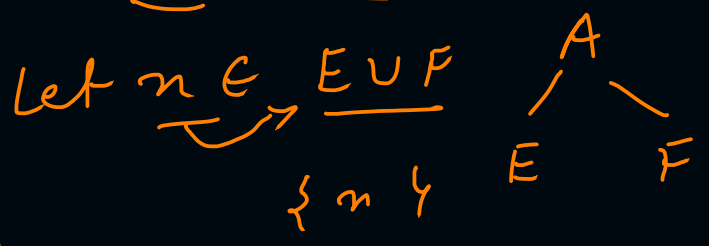
#Q. Let f : A→B be a function, and let E and F be subsets of A. Consider the following statements about images.

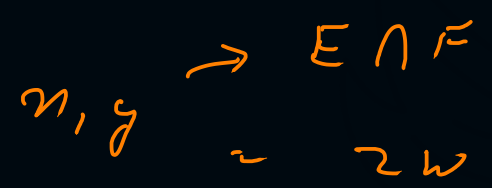
S1: $f(E \cup F) = f(E) \cup f(F)$

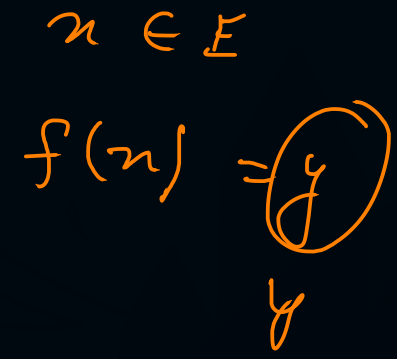
S2: $f(E \cap F) \subseteq f(E) \cap f(F)$

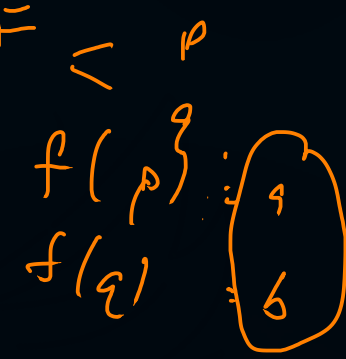
S3: $f(E \cap F) = f(E) \cap f(F)$ if function f is one-one

Which of the following is true?









A. Only S2 and S3 are true.

B. Only S1 and S3 are true.

C. Only S1 and S2 are true.

D. All S1, S2 and S3 are true.

[NAT]

#Q. Let $f(x) = \frac{x}{x+1}$ and $g(x) = \frac{x}{1-x}$ then $(fog)^{-1}(x) = ?$

$fog = f(g(x))$

$y = f\left(\frac{x}{1-x}\right)$

$y = \frac{\frac{x}{1-x}}{\frac{x}{1-x}+1} = \frac{\frac{x}{1-x}}{\frac{x+1-x}{1-x}} = \frac{x}{1-x} \times \frac{1-x}{1}$

$y = x$

x

[MCQ]

#Q. Let $A = R - \{3\}$

$B = R - \{1\}$

Where R is the set of all real numbers if

f: A → B such that $f(x) = \frac{x-2}{x-3}$, then ?

A. f is one-one but not onto

B. f is not one-one but onto

C. f is bijection

D. f is neither one-one nor onto

p, q

$\frac{p-2}{p-3} = \frac{q-2}{q-3} \rightarrow p = q$

$\Rightarrow pq - 3p - 2q + 6$

$= pq - 2p - 3q + 6$

$\Rightarrow q : p$

$\Rightarrow y = \frac{x-2}{x-3}$

$\Rightarrow yx - 3y = x - 2$

$\Rightarrow yx - x = 3y - 2$

$\Rightarrow x = \frac{3y-2}{y-1}$

[MCQ]

#Q. If the function f : $[1, \infty) \to [1, \infty)$ defined by $f(x) = 2^{x(x-1)}$ is invertible, then $f^{-1}(x)$ is ?

A $\frac{1+\sqrt{1+2\log_2 x}}{4}$

B $\frac{1-\sqrt{1+2\log_2 x}}{4}$

C $\frac{1+\sqrt{1+4\log_2 x}}{2}$

D $\frac{1-\sqrt{1+4\log_2 x}}{2}$

$y = 2^{x(x-1)}$

$\Rightarrow \log y = x(x-1)$

$\Rightarrow 0 = x^2 - x - \log y$

$x = \frac{1 \pm \sqrt{1+4\log(y)}}{2}$

[MCQ]

#Q. The domain of the function $f(x) = \sin\left\{\log\left(\frac{\sqrt{4-x^2}}{(1-x)}\right)\right\}$ is?

$N > 0 \quad D > 0$

$\sqrt{4-x^2} > 0 \qquad 1-x > 0$

$\Rightarrow 4-x^2 > 0 \qquad \Rightarrow 1 > x$

$\Rightarrow -2 < x < 2$

$-2 < x < 1$

+ve

$N < 0 \qquad D < 0$

$4-x^2 < 0 \qquad \times$

A. (– 2,0)

B. (– 2,1)

C. (– 2, 2) – {1}

D. (0, 2) – {1}

[MCQ]

#Q. Which of the following statements is /are True ?

A. A constant function is one-one iff domain of the functions has exactly one element

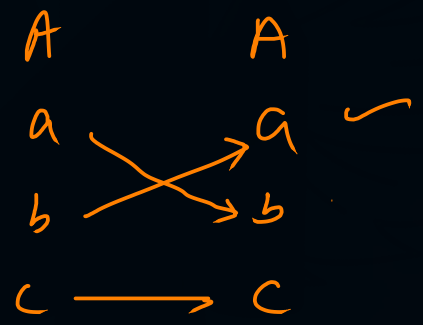
B. A constant function is onto iff co-domain of the functions has exactly one element

C. Every one-one function on a finite set A is bijection

D. Every onto function on a finite set A is bijection.

onto

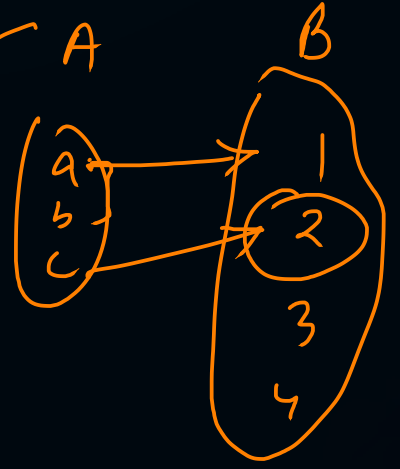
$f(x) = k$



[NAT]

#Q. Let A={a ,b , c} and B= {1,2,3,4}Set C is defined as
C = {f : A → B | 2 ∈ f(x) for some x ∈ A, and f is not one-one}
Then number of elements in set C are ?

functions



Case I image is 2 for one element of A

$^3C_1$ — 3 × 3 = 9

a — 3 —
b x

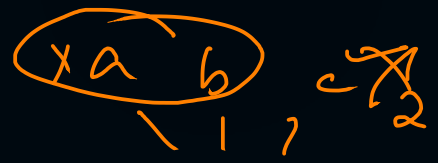
Case II : Two elements

$^3C_2$ = 3 × 3 = 9

Case III

— 1 way

[NAT]



#Q. Let |A|=6 and |B|=3, then number of onto functions possible from set A to set B are_____ ?

Total fun $= 3^6 = 729$ $\quad 1^6 = 1$

$$|S_a \cup S_b \cup S_c| = |S_a| + |S_b| + |S_c| - |S_a \cap S_b| - |S_a \cap S_c| - |S_b \cap S_c| + |S_a \cap S_b \cap S_c|$$

$$= 64 + 64 + 64 - 1 - 1 - 1 + 0$$

$$= 189$$

Onto : $729 - 189 = 540$

THANK - YOU